

Adarsh Bhandary Panambur

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RESEARCH SCIENTIST · AI SYSTEMS · MEDICAL AI · HPC

LLMs/VLMs · Retrieval · Agents · GPU Inference · Clinical Translation

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PROFILE

Research Scientist working across AI systems, high-performance computing, and medical AI. Builds locally deployed and API-backed LLM/VLM applications, retrieval and agent workflows, scalable GPU inference, multimodal foundation models, and clinically validated research software. Experienced in moving from curated datasets and research prototypes to reproducible infrastructure and practical applications.

SYSTEMS EVIDENCE

LOCAL MODELS → RETRIEVAL → AGENTS → GPU INFERENCE → APPLICATION

Local + API-backed runtimes · Grounded context · Tool-using workflows · GPU clusters + SLURM · Research software

PROFESSIONAL EXPERIENCE

Research Scientist · National High Performance Computing Center (NHR@FAU), FAU

06/2026 - Present

Erlangen, Germany

- Design and prototype AI applications using locally deployed LLMs and VLMs, foundation-model APIs, and retrieval pipelines.
- Build tool-using agent workflows and research software with open-source AI frameworks.
- Engineer scalable training, serving, and inference workflows on GPU clusters and supercomputing infrastructure.
- Develop, deploy, and optimize multimodal AI systems for practical research applications.
- Collaborate with academic and industrial partners to translate AI research into practical solutions.

PhD Researcher · Pattern Recognition Lab, FAU

05/2020 - Present

Erlangen, Germany

Thesis: Deep learning algorithms for advanced mammography and breast cancer analysis

- Developed self-supervised learning methods for classification and lesion detection in mammography and contrast-enhanced mammography.
- Built a multi-country mammography repository of approximately 100,000 images for large-scale experimentation.
- Designed and trained multimodal image-text foundation models for breast imaging analysis and radiological reasoning.
- Ranked 4th in the MICCAI ODELIA Challenge 2025 for breast MRI malignancy classification.
- Built reproducible Docker and SLURM pipelines for training, experiment tracking, and large-scale deep learning.

Guest Researcher · West China Hospital, Sichuan University

07/2024 - 09/2024

Chengdu, China

- Validated a Siemens Healthineers breast MRI AI prototype on an external clinical cohort.
- Initiated research to improve breast cancer screening workflows and collaborated with radiologists on annotation quality.

Doctoral Researcher · Siemens Healthineers

05/2020 - 02/2024

Erlangen, Germany

- Validated mammography AI in an international Vietnamese cohort under clinical and regulatory constraints.
- Developed deep-learning methods for mammography positioning-artifact classification and improved annotation workflows with radiologists.

Working Student and Master's Thesis · Siemens Healthineers

10/2017 - 03/2020

Erlangen, Germany

- Built MATLAB workflow prototypes, evaluated IoT and blockchain use cases, and co-invented a secure medical-device unlocking method.
- Master's thesis: CNN-based lung nodule type classification in low-dose CT.

EARLIER MEDICAL-DEVICE ENGINEERING

Senior Engineer · Transcaal Engineers India Pvt. Ltd.	03/2015 - 02/2016
Generated approximately 25% of annual company revenue through QA services across five states; led and mentored a biomedical engineering team.	
Calibration Engineer · Transcaal Engineers India Pvt. Ltd.	09/2014 - 02/2015
Delivered hospital equipment testing, preventive maintenance, and standards-based compliance using Fluke biomedical equipment.	

CAPABILITY MATRIX

MODEL	SCALE	TRANSLATE
Foundation models	GPU computing	Medical imaging
Vision-language models	SLURM and distributed training	Clinical validation
Self-supervised learning	DeepSpeed and Docker	Radiology workflows
Classification, detection, segmentation	Model serving and inference	Research software and reproducible AI
PyTorch · Hugging Face · MONAI · MMDetection · TensorFlow · OpenCV · Linux · Axolotl · SGLang · WandB		

EDUCATION

PhD Research Student · Friedrich-Alexander-Universität Erlangen-Nürnberg	05/2020 - Present
AI / Medical Imaging	
M.Sc. Medical Data Engineering · Friedrich-Alexander-Universität Erlangen-Nürnberg	04/2016 - 03/2020
Data and Image Processing	
B.E. Medical Electronics · Dr. Ambedkar Institute of Technology	08/2010 - 06/2014
Bangalore, India	

SELECTED RESEARCH OUTPUT

- **BE-WISER: Class-Aware Weak Slice-Level Supervision for Breast MRI.** Research Square preprint, 2026.
- **Dataset-Informed Transfer Learning (DITL) for Breast Cancer Screening and Lesion Diagnosis.** CVIP 2025 / arXiv:2607.26043, 2026.
- **Attention-Guided Erasing for Enhanced Transfer Learning in Breast Abnormality Classification.** IJCARS, 2025.
- **MammoBLIP: End-to-End Mammography Report Generation with Vision-Language Models.** IEEE MIC, 2025.
- **BE-WISE: Breast MRI Evaluation with Weakly-Informed Slice-level Explanation.** BVM, 2026.
- **Towards Foundational Models in Mammography: Vision-Language Models for Radiological Reasoning.** Under internal review, 2025.
- **Attention-Guided Erasing for Breast Density Classification.** BVM, 2024.
- **Clinical AI validation studies in Vietnamese mammography and a Chinese breast MRI cohort.** ECR, 2024-2025.

ADDITIONAL

Programming: Python, MATLAB, C++	Languages: English (C2), German (B1), Hindi, Konkani, Kannada
Modalities: Mammography, MRI, low-dose CT, chest X-ray, hematology	Interests: Cooking, padel tennis, hiking, table tennis